Branden Palmer

Active Top Secret/SCI

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Education

University of Texas at Austin | *Austin, TX, USA* **Jan 2026 – May 2029**

- M.S. in Computer Science | GPA: 4.00

Arizona State University | *Tempe, AZ, USA* **Aug 2020 – Dec 2025**

- B.S. in Computer Science | Summa Cum Laude | GPA: 3.83

Open Source Contributions

Plotly.js | *Link* | *Open-source JavaScript charting library behind Plotly and Dash* **Dec 2025 – Aug 2026**

- Co-developed a core **map camera auto-centering and fit-to-data feature**, including geographic bounds handling and edge cases such as **anti-meridian wrapping**, now enabled by default in Plotly.js v4.
- Optimized and benchmarked the implementation for **large point datasets**, while contributing targeted regression tests to ensure correctness across **map subplot configurations**.
- Shipped functionality at the **core-library level**, propagating the capability to downstream Plotly libraries including **Plotly.py (~48M monthly downloads)** and extending its impact across the Plotly ecosystem.

G8 Medical Technology | *Link* | *Medical App* **Jan 2024 – Apr 2024**

Java JavaFX SQLite

- Broke **six-person team** into three feature-focused subteams, **reduced crosstalk**, and increased dev momentum
- Performed **37 merges**, instrumental in synchronizing six-person team
- Delivered **all required specifications on time**

Super Productivity | *Link* | *Productivity App* **Apr 2022 – Jan 2024**

SCSS Angular.js

- ID'd **regression bug** by utilizing **git bisect**
- Corrected UI issue by isolating problematic **SCSS**, resulting change merged into **production branch**

Projects

Job Sourcerer | *Video* | *Link* | *A centralized job application platform* **Jan 2025 – Present**

CD/CI Machine Learning PostgreSQL React SBERT TailwindCSS TypeScript

- Setup **CD/CI pipeline** utilizing GitHub Actions, enabling **early regression detection** and single-click deployment to Firefox and Chrome Extension stores
- Handcrafted **state-of-the-art client-side ATS algorithm** utilizing cosine of similarity and **SBERT semantic matching**, achieved resume to job description matching within **5% of paid service at zero cost**
- Implemented **browser extension architecture** using background, content, and popup scripts to extract job data from third-party sites (e.g., Handshake) and persist **structured records locally**
- Developed a **client-side data layer** using Drizzle ORM with a **PGLite-backed database** to manage jobs, resumes, and derived analytics without a backend dependency

Autonomous Driving Trajectory Prediction | *University of Texas at Austin* | *Deep Learning Project*

Python PyTorch Transformers Tensorboard CNNs NumPy **Jan 2026 – Apr 2026**

- Built and trained **deep learning models in PyTorch** to predict future vehicle trajectories for autonomous driving from lane geometry and raw image inputs
- Developed **model-agnostic training pipelines from scratch**, including data loading, optimization, checkpointing, and offline evaluation
- Evaluated planning accuracy using **longitudinal and lateral trajectory error**, comparing structured-input and end-to-end vision-based driving approaches

Applied Stylometry | *Video* | *Arizona State University* | *Data Science Report* **Nov 2025 – Dec 2025**

Python pandas NumPy scikit-learn NLP TF-IDF PCA SciPy Matplotlib Plotly

- Applied **stylometry and NLP techniques** to analyze authorship signals across multiple text corpora using TF-IDF, character n-grams, and sentence-level features
- Built an **end-to-end text analysis pipeline in Python**, including HTML parsing, text normalization, feature extraction, and exploratory visualization
- Reduced **high-dimensional linguistic feature spaces using PCA** and visualized clustering behavior to compare content-based vs style-based representations
- Evaluated **methodological limitations of stylometry** on short-form texts, demonstrating how data volume and shared background can confound authorship attribution